

Assignment: Laplace Transforms

Part 1: Forward Laplace Transform

Using the Laplace transform table, find $F(s) = \mathcal{L}\{f(t)\}$ for each function below. Assume $f(t) = 0$ for $t < 0$.

1. $f(t) = 4e^{-3t}$
2. $f(t) = 2\sin(5t)$
3. $f(t) = 3\cos(2t)$
4. $f(t) = te^{-2t}$
5. $f(t) = 5 - 5e^{-t}$

Part 2: Inverse Laplace Transform by Partial Fractions

Find $f(t) = \mathcal{L}^{-1}\{F(s)\}$ for each expression using partial fraction expansion. Show all steps.

6. $F(s) = \frac{3}{s(s+3)}$
7. $F(s) = \frac{s+1}{(s+2)(s+4)}$
8. $F(s) = \frac{2}{s^2 + 4s + 3}$
9. $F(s) = \frac{10}{s(s^2 + 4s + 8)}$ (Hint: complete the square in the denominator)

Part 3: Solving ODEs via Laplace Transforms

For each ODE, assume zero initial conditions and a unit step input $r(t) = u(t)$ ($R(s) = 1/s$). (a) Take the Laplace transform. (b) Solve for $C(s)$. (c) Find $c(t)$ via partial fractions and inverse transform.

10. $\dot{c}(t) + 4c(t) = 8r(t)$
11. $\ddot{c}(t) + 5\dot{c}(t) + 6c(t) = 6r(t)$
12. $\ddot{c}(t) + 4\dot{c}(t) + 8c(t) = 8r(t)$ (underdamped: express result in the form $c(t) = K[1 - Ae^{-\sigma t} \cos(\omega_d t - \phi)]$)

Part 4: Pole Locations and Time-Domain Behavior

13. For each transfer function below: identify the poles; state whether the system is stable; and describe the qualitative character of the step response (e.g., overdamped, underdamped, undamped oscillation, or unstable).

(a) $G(s) = \frac{1}{(s+1)(s+3)}$

$$(b) \ G(s) = \frac{1}{s^2 + 2s + 5}$$

$$(c) \ G(s) = \frac{1}{s(s+2)}$$

$$(d) \ G(s) = \frac{1}{(s-1)(s+3)}$$

14. Using MATLAB, create `tf` objects for each system in the previous problem and plot the step responses with `step()`. Confirm that the plots match your qualitative predictions.